

Exploring Anticipatory Head Dynamics for Enhanced Robot Perception in Human–Robot Interaction

Zhichao Chen^{*1} Huthaifa Ahmad^{*2} Liliana Villamar Gómez^{*2} Yuya Okadome^{*3}
 Hiroshi Ishiguro^{*1} Yutaka Nakamura^{*2}

^{*1} The University of Osaka ^{*2} RIKEN ^{*3} Tokyo University of Science

Effective human–robot interaction (HRI) requires robots to exhibit smooth, human-like head dynamics that extend beyond simple reactive control. Existing social robots often orient after stimuli occur, leading to delayed and unnatural motion. Anticipatory behavior, such as proactive head movements based on predicted sound activity, may help maintain conversational flow. However, its benefits in HRI remain insufficiently explored. To investigate this, we conduct a pilot study using Boxie, a minimal robot equipped with essential head degrees of freedom. Participants interact under multiple head-motion strategies in a conversational game. Subjective evaluations indicate that predictive head motion may enhance game experience and perceived social qualities.

1. Introduction

Anticipatory behavior plays a vital role in social interaction [Sebanz and Knoblich(2009)]. Head orientation, in particular, serves as a powerful cue for attention and turn-taking [Fiksdal(1993)]. In group settings, anticipatory head movements often precede speech or gaze shifts, reducing uncertainty [Vertegaal et al.(2003)]. Such movements are tightly linked to embodied social cognition and joint action [Vesper et al.(2010)], blending internal reasoning with outward, socially meaningful signals.

Robots often lack anticipatory capacity, relying instead on reactive control or rule-based. Such methods trigger head movements only after stimuli, producing delayed and socially disconnected behaviors [Mutlu et al.(2009)]. Purely reactive motion fails to convey engagement and diminishes social presence [Huang and Mutlu(2013)]. Few systems employ predictive motion generators [Admoni and Scassellati(2017)], so robot head behaviors typically lack the fluidity and communicative quality of human anticipation.

To explore the influence of anticipatory head motion in human–robot interaction (HRI), we implement multiple head-movement strategies in a conversational game setting. A minimal robot platform is developed to isolate the role of head dynamics. Participants experience game sessions under several motion conditions, including predictive, reactive, and non-informative baselines. After each game, we collect subjective evaluations regarding game experience and robot perception for in-person participants. These measures allow us to analyze how anticipatory head behavior shapes the social and cognitive dynamics of interaction.

Our key contributions are: 1) We develop a minimal robot with teleoperated, anticipatory, and reactive head-motion strategies. 2) We run a pilot conversation study on how head motion shapes game experience and robot perception.

Contact: Zhichao Chen, Osaka University, Graduate School of Engineering Science, chen.zhichao@irl.sys.es.osaka-u.ac.jp

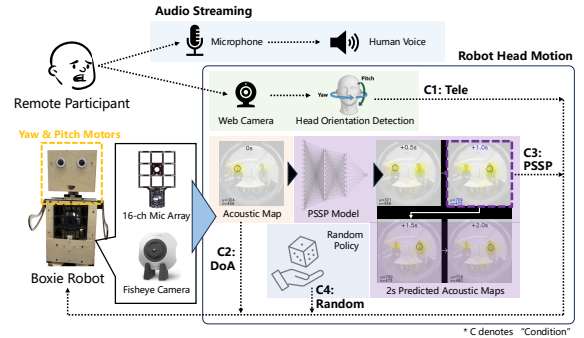


Figure 1: System components of the Boxie robot

3) We present preliminary findings that predictive head motion may enhance perceived social qualities. Given its pilot nature, the study has limitations and calls for deeper follow-up work to support generalizable claims.

2. Word Wolf Game

Participants play the conversation-oriented party game *Word Wolf* to enable natural dialogue during impression evaluations. *Word Wolf* is a simple social-deduction game requiring little prior knowledge, making it effective for eliciting spontaneous conversation.

At the start of the game, each participant receives a hidden word card tied to the topic. Two cards share the same word, and one contains a different word. The citizens—those with the matching word—aim to identify the wolf, who holds the odd word. The wolf tries to avoid detection. All players must infer others’ roles while concealing their own through conversation. Notably, the wolf does not know they are the wolf.

In this study, each group of three participants plays six rounds with different topics and card assignments. An example of a topic is “Skill” and its corresponding two words, for example, “Communication” and “Grit”. Before the game begins, the list of possible topic-related words is

shown to all participants in advance. The details will be described later, but the game is played under various conditions, such as all participants playing face-to-face, or one participant joining remotely through a robot avatar.

3. Boxie Robot

Our goal is to examine anticipatory head dynamics in HRI. To do so, we build a minimal robot, *Boxie*, equipped with two head DoF (yaw, pitch) and two arm DoF.

3.1 Audio Streaming System

We design an audio streaming system in which the robot’s speech is provided by a remote human participant. The setup involves two workspaces: an onsite room for in-person participants and a remote room for remote participants. Each space is equipped with a microphone–speaker pair. Audio from both rooms is streamed via GStreamer to an OvenMediaEngine server and played back through Oven-Player, enabling real-time bidirectional transmission. This ensures that the robot’s speech output directly reflects the live voice of remote participants.

3.2 Head Motion Types

In Figure 1, *Boxie*’s chest has an acoustic camera with a 16-channel microphone array and a fisheye lens. This setup records audio and video, produces acoustic maps, and enables Predictive Sound Source Positioning (*PSSP*) following [Chen et al.(2025)]. To examine how head movements affect human perception, we define four head motions:

- *Teleoperation (Tele)*: A webcam (Figure 2) tracks remote participants’ head orientation, which is used to synchronize *Boxie*’s head movements.
- *Direction-of-Arrival (DoA)*: The acoustic camera generates acoustic maps, from which sound sources can be localized. We extract the pixel with the highest intensity (the sound peak) and convert it into yaw and pitch angles, orienting the robot toward the source.
- *PSSP*: Using the model in [Chen et al.(2025)], acoustic maps are predicted for the next two seconds from the previous five. The +1 s prediction is converted into yaw and pitch angles following the DoA procedure.
- *Random*: As a baseline, yaw angles switch among three zones (left, center, and right) every 10–15 seconds, with random motion within each zone.

4. Game Design

We recruit six students from the Okadome Laboratory at the Tokyo University of Science, dividing them into two groups of three. Each group plays six games, with participants assigned consistent IDs (A/B/C). The experimental setup for each game is shown in Table 1. From Game 2 to Game 6, at least one participant joins remotely, with this role rotating among members. In total, Participants A, B, and C serve remotely 2, 2, and 1 times, respectively.

The experiment includes six conditions: *Onsite*, *Video*, and robot (*Tele* / *PSSP* / *DoA* / *Random*). Example scenes

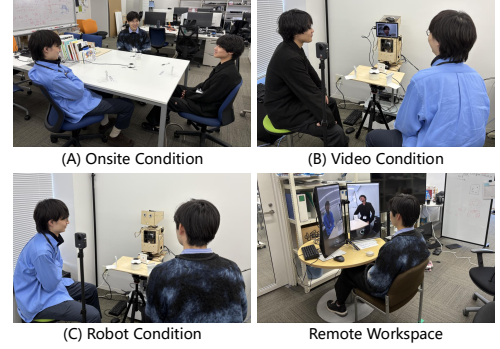


Figure 2: Experimental setup across 3 conditions

Table 1: Three-participant game design

Game No	Role of Participation			Condition
	A	B	C	
1	in-person	in-person	in-person	<i>Onsite</i> (practice)
2	remote	in-person	in-person	<i>Video</i> (practice)
3	in-person	remote	in-person	<i>Tele</i> ¹ / <i>Random</i> ²
4	in-person	in-person	remote	<i>PSSP</i> ¹ / <i>DoA</i> ²
5	remote	in-person	in-person	<i>DoA</i> ¹ / <i>PSSP</i> ²
6	in-person	remote	in-person	<i>Random</i> ¹ / <i>Tele</i> ²

* Condition^x: *x* indicates the group number.

appear in Figure 2. In *Onsite*, all three participants play in person. In *Video*, a small monitor on *Boxie* shows the remote participant’s face while the robot body remains still, mimicking an online meeting. These two conditions serve as practice to familiarize participants with the game and questionnaires before evaluating *Boxie*. From Game 3 to Game 6, all rounds use the robot conditions, with *Boxie* positioned in front of in-person players and executing head motions. Group 1 experiences the motion conditions in the order *Tele* → *PSSP* → *DoA* → *Random*, and Group 2 follows the reverse order.

Each game lasts 3 minutes, followed by a 1-minute break. Including the briefing, six rounds, and post-game questionnaires, each group requires about 1.5 hours.

5. Evaluation Method

At the start of the experiment, we administer a pre-survey to assess participants’ familiarity with the *Word Wolf* game. Those indicating prior knowledge are asked whether they had played before and whether they enjoyed it. This pre-survey captures individual experience and preferences, allowing us to examine how personal familiarity might influence the results. After each game, a post-survey is conducted to address the following research questions: 1) **Wolf?**: Does robot head motion affect game performance? 2) **Tele?**: Can in-person participant identify the robot’s head-movement type? 3) **Game Experience Questionnaire (GEQ)**: How does head movement affect the game experience? 4) **Godspeed Questionnaire Series - Japanese (GQS-J)**: How does the robot’s head movement influence participants’ perceptions of the robot?

Table 2: Sample size of post-game survey

Condition	Role	Wolf? [*]	Tele? [†]	GEQ	GQS
<i>Onsite</i>	in-person	6	-	6	-
<i>Video</i>	in-person	4	-	4	-
	remote	-	-	-	-
<i>Robot</i>	in-person	16	16	16	16
	remote	-	-	-	-

^{*} Wolf? - Who is the wolf in this game? (Options: A / B / C)

[†] Tele? - Is *Boxie* teleoperated? (Options: Yes / No)

Table 2 summarizes the questionnaires and final sample sizes. Participants answer the wolf-guess question and the GEQ [Poels et al.(2007)]; when *Boxie* is present, they also complete the teleoperation-guess question and the GQS-J [Bartneck et al.(2009)]. As no Japanese GEQ translation exists, two native Japanese created one for this study. We use five GEQ subscales relevant to our HRI task: flow, competence, positive affect, negative affect, and tension.

6. Results

6.1 Pre-survey

Six participants complete the pre-survey. All are familiar with *Word Wolf*, and five have played it before; of these, four report enjoying the game. No clear link is found between individual preference and evaluation results.

6.2 Wolf-guess

Table 3: Wolf-guess accuracy by conditions

Condition	<i>Onsite</i>	<i>Video</i>	<i>Tele</i>	<i>PSSP</i>	<i>DoA</i>	<i>Random</i>
Accuracy	50.0%	66.7%	66.7%	83.3%	16.7%	50.0%

Table 3 presents wolf-guess accuracy across conditions. Initial results indicate that conditions may influence accuracy, with *PSSP* yielding the highest result (83.3%) and *DoA* the lowest (16.7%). However, as a pilot study, these results are preliminary and suggest only a potential effect of the conditions on players’ wolf-guess accuracy. A larger sample size is essential to confirm these observations and perform definitive statistical analysis.

6.3 Teleoperation-guess

Table 4: Human control likelihood (in-person)

Condition	Ratio	Yes-Confidence [*]	No-Confidence [†]
<i>Tele</i>	50.0%	50.0	25.0
<i>PSSP</i>	75.0%	100.0	50.0
<i>DoA</i>	50.0%	75.0	25.0
<i>Random</i>	0.0%	-	87.5

^{*} How confident the participant believes *Boxie* is teleoperated.

[†] How confident the participant believes *Boxie* is not teleoperated.

Table 4 summarizes in-person participants’ teleoperation guesses and associated confidence scores, rated from 0

(not at all confident) to 100 (completely confident). The Ratio column shows the proportion of “yes” responses, while confidence ratings (low/medium/high) are converted to 0/50/100 and averaged separately. Results suggest that *PSSP* is associated with the strongest perception of human control, showing the highest agreement ratio (75.0%) and confidence (100.0). *Tele* and *DoA* produce moderate agreement ratios (50.0%) and confidence levels (50.0 and 75.0, respectively). In contrast, *Random* yield no “yes” responses (0.0%), with participants highly confident (87.5) that the robot is not teleoperated. Overall, *PSSP* is most frequently perceived as teleoperated, while *Random* is least teleoperation-like.

6.4 Game Experience

Figure 3 (left) shows GEQ subscale scores across conditions. **Flow** exhibits the clearest differences: *PSSP* shows the highest median score and most consistent responses, potentially indicating that predictive head movement supports engagement. *Tele* also appears to support interaction with relatively high Flow, while *Video* and *Random* register noticeably lower scores, reflecting reduced involvement. *DoA* falls between these extremes.

For **Competence**, *PSSP* again ranks highest, suggesting participants may feel most capable with predictive motion. *Tele*, *DoA*, and *Random* show moderate levels, whereas *Onsite* and *Video* are lowest, possibly highlighting the influence of the robot. For **Positive Affect**, both *PSSP* and *Random* score highly. *PSSP* result appears to align with Flow, while *Random* effect may likely reflect amusement rather than task-related engagement. *Video* shows the lowest positive affect. **Negative Affect** remains low across robot conditions, with *Tele* and *PSSP* lowest. In contrast, *Video* shows the highest values, suggesting less emotional comfort. For **Tension**, *Tele* and *Random* produce the lowest levels. *PSSP* and *DoA* are slightly higher but still below *Video*, which appears to induce the most tension.

Overall, *PSSP* tends to show improvements in Flow, Competence, and Positive Affect while keeping Negative Affect and Tension low. These results suggest predictive head movements may foster the highly engaging interactions, whereas *Video* appears to deliver the weakest experience. *Random* motion seems to boost mood but does not consistently support engagement or performance.

6.5 Human Perception of Robot

Figure 3 (right) shows GQS subscale scores of robot conditions. **Anthropomorphism** is highest for *Tele* and *PSSP*, suggesting that coordinated head movements may enhance human-likeness, while *Random* scores lowest. **Animacy** is similar: *PSSP*, *Tele*, and *DoA* score high, whereas *Random* drops clearly lower, potentially indicating that context-aligned motion conveys greater vitality. For **Likeability**, *Tele* and *PSSP* remain most positive, *DoA* slightly lower, and *Random* highly variable, reflecting mixed impressions. **Perceived Intelligence** is again highest for *Tele* and *PSSP*, moderate for *DoA*, and lowest for *Random*, consistent with the idea that goal-directed motion signals intelligence. **Perceived Safety** shows little variation

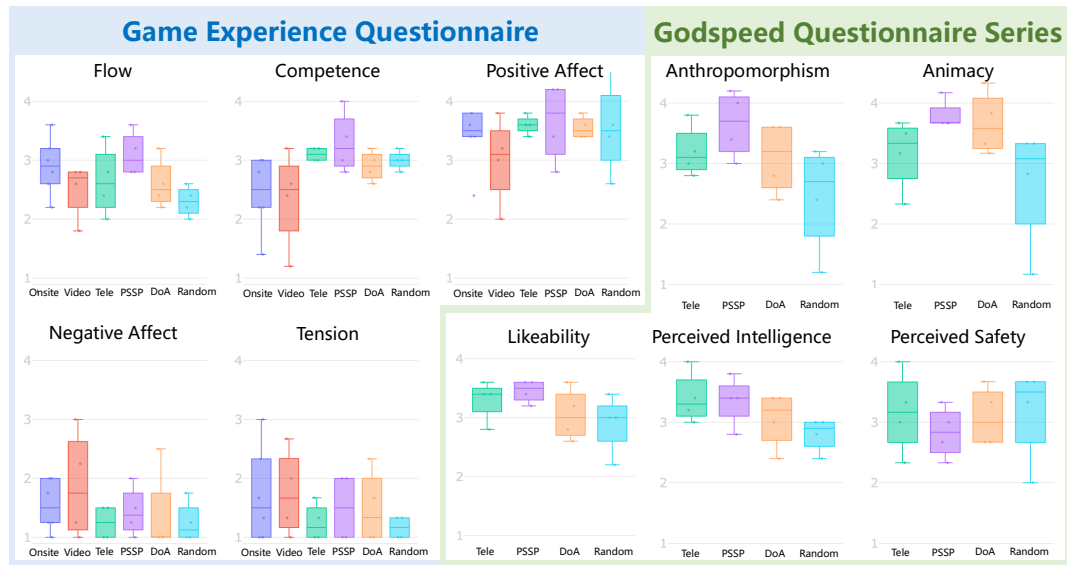


Figure 3: In-person GEQ and in-person GQS results across conditions

across conditions, with no strong negative ratings.

Overall, *PSSP* is associated with the strongest evaluations in anthropomorphism, intelligence, and likeability, with *Tele* close behind. *DoA* occupies a middle ground, while *Random* motion ranks lowest across most subscales, underscoring the importance of context-aware behavior.

7. Conclusion

This pilot study provides an initial examination of how anticipatory robot head movements influence interaction experience and robot perception. Given the limited sample and design constraints, further research is needed to validate and generalize these results.

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